

WHestBench Canonical Research State

ARC White-Box Estimation Challenge 2026

Full reconciliation of experiments, evidence quality, closure scope, and next decisions

Canonical policy: a small or synthetic sandbox result may close the exact implementation it tested, but it does not close a broad research avenue unless the environment is matched, the evidence is replicated, or a rigorous theorem applies.

Evidence cutoff: July 29, 2026, 12:36 p.m. ET

Companion workbook: [whestbench_canonical_research_ledger_20260729_reconciled.xlsx](#)

1. Executive summary

The research has localized a large, real, late-layer correction channel, and the adjoint compute problem is largely solved. The unresolved problem is observability: cheaply predicting the signed absolute center/source defect that the complete Kerdock rule misses.

- The only shipping-ready direction is arithmetic: compare the fixed three-layer and adaptive depth-2–6 suffix compilers inside the current partial-tree implementation with measured FlopScope and residual wall time.
- The strongest validated mechanism is the layer-31 post-ReLU mean defect: exact correction produced candidate/base 0.2187 on 64 holdout networks with 64/64 wins.
- Connected-cubic controls contain transferable signal. Oracle-state rank-4 c21 reached candidate/base 0.5993 on 16/16 networks; the M85 factorized-anchor control reached 0.8814 on frozen holdout but had a 1.4277 worst-network tail.
- The checkpoint adjoint handoff is no longer merely a derivation. A local layer-24 q=32 implementation reproduces the factorized target within 2.31% at about 10.3B compute; however, it is still roughly 31% from the oracle anchor.
- The 128-probe 0.190 candidate/base claim is still only an eight-network development ceiling. The frozen probes, arrays, and untouched output are missing from shared artifacts.
- Several attractive closures were recorded too broadly. Network-specific K2 coresets, conditional integration, stable-gate MLMC, ordinary sampled pilots, and reduced-width equivariant models are negative for their exact tested forms, but not all imply class-level impossibility.

What changed in this reconciliation

- Added 18 missing or overly compressed experiment rows, including the local adjoint implementation, oracle-state rank-4 c21, source-closure failures, granular activation-region experiments, and individual equivariant-model results.
- Corrected stale or wrong top-level statements: the old 1.442e-7 compiler projection used an obsolete baseline; the exact quadratic control is closed; the stable-gate MLMC report did not contain the previously attributed 2.02 worst-network value.
- Downgraded the claim that sparse centering is low-dimensional. It is exactly shared, but its empirical rank on the real frozen probes is unknown; a width-256 proxy span has rank 128.
- Inserted the missing negative toy evidence against short-suffix source localization. Real source attribution is now a prerequisite for micro-cubature or suffix-only closure work.
- Separated tested-form closure from broad-class closure throughout the ledger.

2. Evidence hierarchy and closure rules

Tier	Meaning	Allowed conclusion
E5 — official paired grader	Exact benchmark, current implementation, target comparison, full compute/wall accounting	Submission decision and measured score claim
E4 — frozen full-width holdout	Fresh width-256/depth-32 networks after settings are frozen	Primary pre-official evidence; may close a tested implementation
E3 — architecture-matched synthetic	Correct width/geometry but synthetic targets, development use, or incomplete runtime fidelity	Select or kill exact variants; do not make broad class claims
E2 — oracle mechanism	True moments, target-aware directions, unavailable anchors or interventions	Establishes headroom and structure only
E1 — algebra / infrastructure	Identity proof, unit test, code harness, complexity estimate	No MSE or deployment claim

Mandatory provenance fields

- Execution environment: local full repository, exact code archive, architecture-matched sandbox, wrong-width sandbox, or official runner.
- Network architecture and geometry: width, depth, exact Kerdock point count, basis count, support size, and whether the real seed-3 design was used.
- Stage and sample scale: screen, frozen validation, holdout, training/calibration sizes, network IDs or seed manifest, and reference-target precision.
- Estimator status: oracle, deployable-shaped, fully deployable, infrastructure-only, or projected.
- Performance convention: candidate/base lower-is-better or baseline/candidate gain higher-is-better; raw MSE and adjusted score must be separated.
- Risk: wins, worst network, confidence interval, reference-noise fraction, compute estimate versus measured cost, and protected-holdout use.

Governance correction: global IDs 0–199 are exposed. The V73 64-network block is valuable oracle-mechanism evidence, but it is not available as a fresh deployable-estimator holdout.

3. Canonical portfolio

Branch	Status	Best evidence	Scale / caveat	Next decision
Shipping compiler	Shipping pending measurement	Adaptive calibrated adjusted ratio 0.97865 on 60 fresh networks; fixed3 is simpler	Final integrated current-package cost not measured; old absolute projection is stale	Paired partial-tree baseline vs fixed3 vs adaptive
Layer-31 correction	Leading oracle target	Full oracle candidate/base 0.2187, 64/64; sparse K12 gain 1.7844 and K32 2.9565	Sparse support is target-aware N=8; signed deployable coefficient missing	Freeze target-free support and independent sign estimator
Sparse cubic	Leading unvalidated ceiling	128-probe 0.190 claim on N=8; oracle-state rank4 c21 0.5993 on 16/16	Frozen sparse assets missing; anchor unavailable	New exact-anchor validation, then center/cubic ablation
Adjoint K3	Implementation validated, source open	q32 layer24: 2.31% factorized error at ~10.3B; q64: 0.117% at 13.34B	q32 remains ~31% from oracle; source closures failed	Real source-layer attribution and independently centered source
Micro-cubature	Open bounded after attribution	Synthetic externally centered transport only	Toy short-suffix localization was unfavorable; real M83 tolerance much stricter	One frozen configuration only after real rank/localization
Companion phase design	Open adjacent theory	Independent pilots become stable only when compute erases benefit	No shared-arithmetic implementation exists	Second phase estimate with <1–2% true incremental compute
Equivariant learning	Paused / one bounded width-256 test	Best reduced-width phase encoder gain 1.01142; replay break-even 1.03125	Wrong width, tail 6.12; broad models narrowed, not universally closed	Grouped width-256 edge-DWS after target freeze
Network-specific coreset	K2 tested form closed; broad class paused	Oracle support 2.391e-9 added MSE; deployable K2 1.282e-7	Exact geometry synthetic N=20; protected holdouts unopened	New proxy-to-oracle support evidence or matched stronger local method
Activation-region / phase stats	Closed tested forms	Residual CV candidate/base 1.0112 on frozen 24; learned phase statistics ~1.000	Sparse-coordinate oracle is retained	No more orthant/gate tuning
Stable-gate MLMC	Closed statistically	Candidate/base 1.2645 despite correlation 0.99944	Partial allocation destroyed complete-design cancellation	Compiler arithmetic only
Harmonic controls	Closed standalone	Broad frozen candidate/base 1.00444; q8 frozen 0.995, worst 1.471	May remain an internal exact-zero-mean feature	No standalone revival

4. Reconciliation audit: what was missing or misrecorded

Issue	Previous / ambiguous recording	Corrected evidence	Environment / scale	Confidence	Canonical correction	Source
R1 Compiler absolute projection	Adaptive compiler listed as $\sim 1.442e-7$ best near-term score.	$1.442e-7$ was calibrated from older $1.473e-7$ assembly-free baseline; current partial-tree row0 is $\sim 1.12e-7$ but not full-suite. Relative 0.97865 remains valid evidence.	Local N=60 plus row0 checkpoint	High	Use relative ratio only; run integrated paired measurement.	NONK3_CONTINUATION_REPORT; compendium
R2 Quadratic control stale open	Current State still called 4–7% exact quadratic best open non-K3 control.	Fresh frozen N=12 reversed to candidate/base 1.0188.	Local full-width N=12	High	Closed.	Ledger A31 / current-state history
R3 M85 tail omitted	Shrunk worst network listed as not reported.	70% blend worst candidate/base 1.4277; unshrunk worst 1.8312.	Local frozen N=8	High	Record tail; mechanism is real but unsafe.	arc_code deep audit
R4 Adjoint understated	Recorded as analytic/sandbox proposal with 20B/<10B estimate.	Local q32 layer24 implementation has 2.31% factorized error at $\sim 10.34B$; q64 0.117% at 13.34B. Still $\sim 31\%$ from oracle.	Local frozen 160–167	High	Mark implementation validated against factorized anchor; closure remains open.	adjoint_k3_full_holdout8.json
R5 Source failures omitted	No canonical row for local closure/source insertion failures.	Born/Gaussian $\sim 18\%$ factorized/ $\sim 38\%$ oracle; checkpoint sample-only 1.71/1.69; direct insertion catastrophic.	Local archive, limited blocks	Moderate-high	Add M96 and close exact tested sources.	arc_code deep audit
R6 Strong oracle c21 omitted	Sparse/cubic narrative centered on N=8 0.190 claim.	Rank4 oracle-state c21 has candidate/base 0.5993 on 16/16, worst 0.9835.	Local full-width N=16 oracle	High mechanism	Add as stronger replicated feature/control evidence.	arc_code deep audit
R7 Shared vs low-rank	Canonical document said sparse center map was shared low-dimensional.	Shared at most 256D is exact; toy rank 90=6 is synthetic, while width 256 proxy span rank 128. Real rank unknown.	Analytic+toy/proxy	High correction	Replace 'low-dimensional' with 'shared, empirical rank unknown'.	sparse center report; terminal test
R8 Short-suffix optimism	Adjoint/micro-cubature described short suffix 5–10B as promising without negative localization evidence.	Toy source localization retained $\sim 4.7\%$ gain at depth 2–4 and was harmful at 6–16; full depth required.	Synthetic unit test	Moderate warning	Real source attribution becomes prerequisite; do not assume locality.	agent2_sparse_adjoint_results
R9 Stable MLMC worst	Evidence Registry attributed worst 2.02 to stable-gate MLMC.	2.02 belongs older V70 multilevel test; V77 report gives aggregate 1.2645, median 1.5077, no worst.	Sandbox exact-design proxy	High	Correct worst field to not reported.	FULL_MULTILEVEL report

Issue	Previous / ambiguous recording	Corrected evidence	Environment / scale	Confidence	Canonical correction	Source
R10 Partial-MUB nuance	Recorded mainly as closed extra-trajectory pilot.	N=8 16-basis result is uncertain/tail-unsafe, but 64/128 bases make all networks improve; compute erases benefit. This proves sign is estimable with enough independent samples.	Fresh sandbox N=8	Moderate	Close ordinary pilots but open shared-arithmetic companion information.	FRESH_SCREEN / continuation
R11 Activation experiments compressed	Many distinct phase and pilot experiments were merged into one V75 row.	Conditional CV, Walsh learning, invariant contractions, rotated pilots and internal pilots have distinct scales/results/reopening implications.	N=8/24 and larger corpora	High	Add ARCI-4 through ARCI-9 rows.	Activation continuation
R12 Equivariant experiments compressed	Only one aggregate M92 row.	Eight architectures/probes at widths 32/64 show different tails; best 1.01142 still below replay break-even. Three zero-target rows were repaired.	Large reduced-width corpora	High	Add M97–M103 and A38; preserve one bounded width 256 test.	Equivariant report
R13 Sparse toy tolerance conflict	Toy sparse center errors up to ~1.7% did not hurt, which could be misread as relaxed tolerance.	Real M83 breaks ~0.45% and 0.65% is harmful. Toy is algebra/unit test only.	Toy vs real oracle tolerance	High	Never use toy tolerance for ARC gates.	sparse center report / M83
R14 Coresets closure scope	Sandbox agent recommended broad closure; user/local work not finished.	K2 exact tested representation fails decisively, but oracle supports exist and protected holdouts were unopened.	Exact geometry synthetic N=20	High K2 / low broad class	Close tested K2 only; broad class paused until qualitatively new proxy or matched local replication.	Exact Kerdock coreset report
R15 Archive completeness	Code archive could be mistaken for canonical complete state.	Archive omits official data, high-precision corpora, frozen sparse arrays and sibling continuations.	655-file archive audit	High	Treat archive as code/result snapshot, not complete ledger.	arc_code deep audit

5. Detailed findings by research family

5.1 Shipping compiler and benchmark context

- Adaptive depth 2–6: raw candidate/base 1.00170; calibrated adjusted ratio 0.97865 [0.97566, 0.98163]; 59/60 calibrated wins; worst 1.00222.
- Fixed three-layer package: 2,064-row basis-balanced pilot, final three layers, rare-switch threshold 8, middle correction 0, final shrinkage 0.875, and a full-suffix fallback below 0.5% predicted savings.
- The often-quoted $\sim 1.442\text{e-}7$ adaptive score was derived from the older $1.473\text{e-}7$ assembly-free baseline. The current partial-tree row-0 projection is $\sim 1.12\text{e-}7$, but is not a paired full-suite score. The relative compiler ratio is the portable result.

Decision: measure before further research. Packing, symbolic composition, residual wall time, and fallback frequency can determine the winner.

5.2 Layer-31 sparsity and signed coefficient estimation

Coordinates	Raw gain	Oracle benefit captured	Worst candidate/base
4	1.2991x	29.83%	0.8693
8	1.5612x	46.15%	0.7371
12	1.7844x	55.23%	0.6607
16	2.0597x	63.99%	0.5870
32	2.9565x	81.39%	0.4351

- Observable Gaussian-gap $\times$ final-sensitivity selection achieved 1.1795x at K=12 and 1.3645x at K=24 when exact signed corrections were granted. Coordinate selection is not the main obstacle.
- The partial-MUB experiment confirms that independent samples do contain the sign. Sixteen bases had a strong but unsafe N=8 point estimate; 64 and 128 bases made all networks improve, but computation consumed or exceeded the gain.
- The final-ReLU linearization is essentially exact in the rank-12 pilot study. The failure is coefficient estimation, not final replay or mode capacity.

5.3 Sparse cubic controls and adjoint implementation

- The exact sparse recentering identity is shared: all probe center errors can be written as $B\delta + Q(\delta^2)$. This eliminates the idea of 128 independent mean anchors.
- It does not prove a small target. The toy $d=20$ map had rank90=6, while a width-256 proxy terminal span had rank 128. The real frozen probe map must be measured.
- The batched adjoint exactness tests pass to floating precision. Shared pullbacks are cheap; source generation and projection dominate.
- A local factorized-anchor checkpoint implementation already reaches $\sim 10.3\text{B}$ at $q=32/\text{layer}24$, but it accurately reproduces a factorized target that is still $\sim 31\%$ from the oracle.
- Toy source localization argues against assuming a short suffix: depths 2–4 retained only $\sim 4.7\%$ of exact gain, and depths 6–16 were harmful.

Decision: recover the real frozen probes, measure B rank and layerwise source attribution, then test one independently centered source. Do not spend time on more backward algebra.

5.4 Coresets and particle compression

Arm	Mean added MSE	Worst	Networks $\leq 2.2e-8$
Random + anchor weights	2.069e-7	4.673e-7	0/20
Anchor-kernel coreset	1.341e-7	2.781e-7	0/20
Deployable suffix K2	1.282e-7	2.797e-7	0/20
K2 support + oracle weights	2.941e-8	1.646e-7	16/20
Oracle support + oracle weights	2.391e-9	1.496e-8	20/20

- The exact tested K2 feature/selection/calibration family is strongly negative. However, oracle support proves that compression itself is possible.
- The correct reopening test is not another optimizer sweep. A qualitatively new representation must first predict oracle row importance or same-support oracle error on fresh networks.
- Because this was synthetic $N=20$ in a sandbox and stronger local work may differ, the broad network-specific coreset class remains paused rather than universally closed.

5.5 Activation regions, phase statistics, and companion designs

Experiment	Frozen result	Conclusion
Conditional-region residual CV	Gain 0.9889; 8/24; worst candidate/base 1.195	Predicted correction is essentially orthogonal to true Kerdock defect
Walsh/basis learning	Best of 96 models 1.00056 optimistic	Earlier ~4% point estimate was small-sample instability
Invariant phase statistics	Best 1.00092; worst 1.0266	No useful cubic/quintic/PCA-skew statistic in tested family
Independent 1,032-point pilot	Gain 0.9296; one network +81.1% MSE	Unstable
Independent 2,064-point pilot	Raw gain 1.0394; optimistic adjusted ~0.9912	Less than 1% headroom; no decisive interval
Internal 516-row pilot	Gain 0.9651; 7/24	Same-cloud disagreement is not a phase estimate

Opposite-direction insight: independent phase information becomes more useful as more points are spent. The promising theory is therefore a companion cubature that reuses nearly all prefix arithmetic—not a better orthant solver or ordinary extra trajectories.

5.6 Equivariant learning

Experiment	Validation scale	Replay gain	95% interval	Worst candidate/base
Forward target-31, width32	64	1.00235	0.99989–1.00493	1.55
Bidirectional, width32	64	1.00287	0.99048–1.00896	9.85
Forward target-31, width64	32	1.00300	0.99166–1.01773	1.08
Bidirectional, width64	32	0.98933	0.96033–1.03495	1.23
Large nonlinear basis probe	128	1.00815	0.94623–1.05560	4.62
Phase-augmented basis, sorted	128	0.96380	0.82991–1.07185	16.62
Phase-augmented weight encoder	128	1.01142	0.98266–1.03259	6.12

- One final replay increases effective compute by 3.125%, so a free model still needs at least 1.03125x raw gain to break even.
- The reduced-width campaign tested target redesign, bidirectionality, larger corpora, full basis diagnostics, and exact phase augmentation. None cleared the gate.
- The scientific status is narrowed, not closed. The only defensible continuation is a grouped width-256 edge-state Deep Weight Space model trained through the scored final replay and stopped unless grouped CV gain is at least 1.10x with controlled tails.

6. Opposite results that suggest better experiments

Opposite results / tension	Synthesis	Best future experiment	Hard gate	Evidence caution
Oracle support nearly exact vs deployable coreset fails 11.7x gate	Coreset existence is not the issue; observable row-importance representation is.	Before optimization, regress/measure a cheap proxy against oracle row importance/support recall on fresh networks.	Proxy must predict support with same-support oracle added MSE $\leq 1.1e-8$ and safe worst case before any solver sweep.	Exact geometry but synthetic N=20; broad class remains open only qualitatively.
Layer31 oracle huge vs every phase predictor near zero	Coordinates and downstream map are accessible; signed absolute defect is missing.	Independently centered target-contracted source estimator or companion cubature; evaluate signed coefficient cosine and final replay.	$\geq 1.3x$ raw on frozen validation, positive cosine, safe tails, positive adjusted score.	Do not confuse activation-distribution accuracy with cubature-phase accuracy.
Partial-MUB sign improves with sample size vs adjusted score collapses	Independent samples contain the sign, but ordinary trajectories cost too much.	Shared-arithmetic companion rule reusing first31 layers / basis transforms; target only 8–16 coefficients.	True incremental compute $\leq 1-2\%$, raw gain ≥ 1.10 or strong sparse-control composition.	N=8 scaling is suggestive, not a validation claim.
Oracle rank4 c21/M85 gains vs factorized/source closure failures	Control features are useful; anchor approximation is the sole bottleneck.	Freeze features/regression and compare independent center/source estimators with oracle/deployable ablations.	Retain $\geq 70-90\%$ oracle-control gain under $< 10-14B$ and safe worst network.	Do not reopen feature searches before anchor test.
Adjoint factorized handoff exact/cheap vs 31% oracle gap	Backward contraction solved the wrong target accurately; factorized closure omits real source.	Measure signed layerwise oracle source decomposition and design only target-contracted local source corrections.	Real source model must reduce factorized-to-oracle gap materially and improve complete-control MSE.	Factorized-anchor accuracy is not oracle-anchor accuracy.
Shared center map toy rank6 vs width256 proxy rank128	Shared does not imply compressible; real rank may be high.	Recover frozen probes and compute real B singular spectrum, downstream weighting and layer31 overlap.	Proceed to rank ≤ 16 predictor only if $\geq 90\%$ relevant energy retained.	Toy algebra is not empirical ARC evidence.
Stable-gate surrogate correlation 0.99944 vs MLMC 26.5% worse	Pointwise/ R^2 and i.i.d. block variance ignore deterministic global Kerdock cancellation.	Any multilevel estimator must preserve complete designs as units or use algebraically zero-mean corrections.	No arbitrary partial-basis allocation; full coupled MSE must improve after cost.	High correlation is a misleading surrogate metric.
Degree6 is first live harmonic vs harmonic controls fail	First live does not mean dominant or observable; ensemble degree6 ceiling only $\sim 13.9\%$.	Use degree6 only as an exact-zero-mean internal feature for an independently anchored late control.	No standalone revival without $\geq 5\%$ adjusted frozen gain and safe tails.	Two separate frozen failures exist.
Mean-field bias learnable vs Kerdock phase not learnable	Smooth closure error is predictable; orientation-sensitive high-degree phase is not with tested features.	Predict residual coefficients of an independent analytic/micro-cubature anchor, not raw Kerdock residual.	Width256 grouped CV gain ≥ 1.10 , lower CI ≥ 1.05 , tails controlled.	Reduced-width negatives narrow but do not universally close DWS.
Oracle x1/x1a recursive gains vs PSD clipping destroys them	The useful correction may live in contracted outputs even when no coherent full covariance state exists.	Propagate only adjoint-weighted contractions or learn a valid factorized state intrinsically; avoid post-hoc clipping.	Complete estimator must retain gain without repeated repairs.	Oracle features and invalid state are mechanism evidence only.
Same-cloud Edgeworth expected near-neutral vs observed 12.996	Same-cloud corrections can actively corrupt the binding mean, not just telescope.	Use external absolute center and test mean-only ablation before any cubic correction.	Mean error $< 0.3\%$, break-even $< 0.45\%$, complete-control tail safe.	Predictions from tensor accuracy were wrong; rely on measured final MSE.

Opposite results / tension	Synthesis	Best future experiment	Hard gate	Evidence caution
Small screens produce large leads vs frozen reversals	Network heterogeneity and target noise dominate many sub-5% effects.	Immutable 8/24/64+ stages, network bootstrap, reference-noise correction and no holdout reuse.	Major path $\geq 1.3\times$ screen; cheap additive $\geq 5\%$ adjusted with CI and tail gates.	Small synthetic evidence can kill exact code paths but should not close broad classes.

7. Canonical next experiments

Priority	Experiment	Precondition	Gate
1	Integrated compiler measurement	Current partial-tree baseline and exact fixed3/adaptive implementations	Measured adjusted improvement; no runtime/prediction regression
2	Frozen sparse exact-anchor validation	Recover and hash layer/probes/folds/regularization/shrinkage; new network block	Candidate/base ≤ 0.50 , preferred ≤ 0.30 ; safe tails
3	Real sparse-map rank and channel overlap	Frozen probes and exact layer31 replay corrections	Rank ≤ 16 retains $\geq 90\%$ downstream-relevant correction energy before low-dimensional learning
4	Real layerwise source attribution	Implemented q32/q64 adjoint and independent oracle/factorized targets	Short suffix only if ≤ 12 layers retain $\geq 90\%$ control gain
5	One independent contracted source / micro-cubature	Attribution identifies target span and checkpoint	Complete-control ≤ 0.60 ; real mean tolerance; $\leq 8-10B$; safe tails
6	Shared-arithmetic companion phase rule	Reuse nearly all protected prefix computation	$\leq 1-2\%$ true incremental compute and positive signed coefficient signal
7	Target-free 8-32 coordinate selector	New networks, no signed truth in selector	K ≤ 32 captures $\geq 50\%$ oracle gap with stable support
8	Bounded width256 edge-DWS	Independent anchor and basis frozen	Grouped CV gain ≥ 1.10 ; lower CI ≥ 1.05 ; worst ≤ 1.25 ; adjusted positive
9	Qualitatively new coreset representation	New proxy demonstrates oracle-row importance correlation	Same-support oracle added MSE $\leq 1.1e-8$ before optimization

What not to spend time on

- Larger orthant solvers, more explicit uncertain gates, or deeper activation-region enumeration.
- More stable-gate MLMC allocation over partial Kerdock bases.
- More q sweeps, exchange rounds, signed-weight sweeps, or pilot output metrics for the tested K2 coreset.
- More same-cloud Edgeworth, delete-fold, or internal-subdesign anchors.
- More ordinary independent trajectories for layer-31 sign estimation unless arithmetic is shared.
- Standalone degree-6/harmonic regressions or broad reduced-width weight models.
- Any new statistical claim using IDs 0-199 or a previously opened branch holdout.

Appendix A. Key source artifacts reviewed

- WHestBench research catalogs and experiment ledgers (multiple July 29 versions)
- arc_code.zip and arc_code_deep_audit_20260729.md
- NONK3_CONTINUATION_REPORT.md and compiler implementation handoffs
- ARC_ACTIVATION_REGION_CONTINUATION_REPORT.md
- FULL_MULTILEVEL_PREFIX_SUFFIX_REPORT.md
- NETWORK_SPECIFIC_KERNEL_CORESET_EXACT_KERDOCK_REPORT.md
- EQUIVARIANT_WEIGHT_MODEL_RESEARCH_20260729.md
- Independent Short-Suffix Micro-Cubature for Sparse Late-Layer Anchors
- AGENT2_EXACT_SPARSE_ADJOINT_REPORT.md and agent2_sparse_adjoint_results.json
- sparse_cubic_center_and_channel_report.md and agent2_agent1_terminal_test.json
- FRESH_SCREEN_SUMMARY.json and AGENT1_CONTINUATION_SUMMARY.json
- adjoint_k3_full_holdout8.json and M85/connected-cubic result JSONs
- ARC_HARMONIC_CV_EXPERIMENT_20260729.md
- Agent 7 audit and split-contamination report

Appendix B. Ratio convention

Canonical workbook convention: candidate MSE / baseline MSE, so lower is better. When a source reports a gain as baseline/candidate, the document labels it explicitly as a gain or converts it before comparison.

Appendix C. Scope of canonicity

The companion workbook is the exhaustive experiment tracker. This narrative document is the canonical decision summary. The source library still contains raw scripts, archives, and result files that may not be reproducible in a sandbox because official data, high-precision moment corpora, or frozen arrays are absent. Absence from a code archive is not evidence that an experiment was never run.